

A Partial Bound for the Midpoint Constant of Near-Isometries

Run fa_banach_001, agent_lane_10

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Source question

The source is Rafal Gorak, *Perturbations of isometries between Banach spaces*, arXiv:1105.0854. In Section 4, the paper defines, for $\mu(t) = (1 + \varepsilon)t$,

$$K_\varepsilon = \sup \frac{\|T(\frac{a+b}{2}) - \frac{Ta+Tb}{2}\|}{\varepsilon \|a-b\|},$$

where the supremum is over all μ -isometries T between Banach spaces and all $a \neq b$ in the domain of T . The open constant is

$$K = \limsup_{\varepsilon \rightarrow 0} K_\varepsilon.$$

The paper records $K \geq 0.5$, $K \leq 3$, and asks for the exact value.

$\varepsilon \rightarrow 0$. We have already seen that $\|T(\frac{a+b}{2}) - \frac{Ta+Tb}{2}\| = O(\varepsilon \|a-b\|)$ as $\varepsilon \rightarrow 0$. It is very easy to show that this is everything one can obtain in the general case. Indeed, as Vestfrid noticed in [11], consider $T : \mathbb{R} \mapsto \mathbb{R}$ defined as:

$$T(x) = \begin{cases} (1 + \varepsilon)x & \text{if } x \geq 0 \\ \frac{1}{1+\varepsilon}x & \text{if } x < 0. \end{cases}$$

However the exact value of the constant bellow remains unknown:

$$K = \limsup_{\varepsilon \rightarrow 0} K_\varepsilon$$

where

$$K_\varepsilon = \sup \frac{\|T(\frac{a+b}{2}) - \frac{Ta+Tb}{2}\|}{\varepsilon \|a-b\|}.$$

Supremum is taken over all T - μ -isometries between Banach spaces, where $\mu(t) = (1 + \varepsilon)t$, and over all pairs of points $a \neq b$ from the domain of T . The above example shows that $K \geq 0.5$ and Corollary 3.3 shows that $K \leq 3$. It is worth to notice that a simple analysis of the proof of Corollary 3.3 gives us that $\liminf_{\varepsilon \rightarrow 0} K_\varepsilon \leq e$.

Finally it is of a great interest to find the optimal constant M_0 and the optimal estimation of δ in Theorem 1.4. In particular it is still unknown whether the constant $M_0 = 1/\sqrt{2}$ is the optimal one or not. However we skip

Result

Theorem 1. *For Gorak's midpoint constant,*

$$1 \leq K \leq \frac{e^{\sqrt{2}}}{\sqrt{2}} < 3.$$

Idea of the proof

The source lower bound uses a one-dimensional map with different slopes on the two half-lines. A two-dimensional shear gives twice the asymptotic midpoint defect for the same first-order bi-Lipschitz distortion. The map is still globally controlled because each difference is governed by a shear matrix of the form

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$$

with $|a|$ bounded by the same parameter. For the upper bound, no new midpoint theorem is needed: optimizing the dyadic parameter already present in Gorak's Theorem 2.1 gives a uniform asymptotic constant below 3.

Proof

Let $M = 1 + \varepsilon > 1$ and put

$$\eta = M - M^{-1}.$$

On the Euclidean plane $\mathbb{R}^2$, define

$$T(s, t) = (s + \eta|t|, t).$$

This map is bijective, with inverse $(u, v) \mapsto (u - \eta|v|, v)$.

Lemma 1. *T is M -bi-Lipschitz.*

Proof. For two points (s, t) and (s', t') , write

$$u = s - s', \quad v = t - t'.$$

If $v \neq 0$, set

$$\theta = \frac{|t| - |t'|}{t - t'},$$

and if $v = 0$, set $\theta = 0$. Then $|\theta| \leq 1$ and

$$T(s, t) - T(s', t') = (u + \eta\theta v, v).$$

Thus this difference is obtained from (u, v) by the matrix

$$A_a = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, \quad a = \eta\theta, \quad |a| \leq \eta.$$

The Euclidean operator norm of A_a is

$$\|A_a\| = \frac{\sqrt{a^2 + 4} + |a|}{2}.$$

Since this expression is increasing in $|a|$ and

$$\frac{\sqrt{\eta^2 + 4} + \eta}{2} = M \quad \text{when} \quad \eta = M - M^{-1},$$

we get $\|A_a\| \leq M$. Also $\det A_a = 1$, so the least singular value of A_a is at least M^{-1} . Therefore

$$M^{-1} \|(u, v)\|_2 \leq \|T(s, t) - T(s', t')\|_2 \leq M \|(u, v)\|_2.$$

□

Hence T is a μ -isometry for $\mu(r) = Mr$. Now take

$$a = (0, 1), \quad b = (0, -1).$$

Then $(a + b)/2 = 0$, $T(0) = 0$, and

$$\frac{Ta + Tb}{2} = (\eta, 0).$$

Since $\|a - b\| = 2$, this example gives

$$K_\varepsilon \geq \frac{\eta}{2\varepsilon} = \frac{M - M^{-1}}{2(M - 1)} = \frac{M + 1}{2M}.$$

Letting $M \downarrow 1$, equivalently $\varepsilon \downarrow 0$, gives $K \geq 1$.

For the upper bound, apply Gorak's Theorem 2.1 to $\mu(r) = Mr$. With $k = 2^{n+1}$, it gives

$$\left\| T\left(\frac{a+b}{2}\right) - \frac{Ta + Tb}{2} \right\| \leq M^{k-1} \frac{\|a - b\|}{k}.$$

Therefore

$$K_\varepsilon \leq \inf_{k \in \{2, 4, 8, \dots\}} \frac{M^{k-1}}{\varepsilon k}.$$

Let $h = \log M$. For small ε , choose a dyadic k such that

$$\frac{1}{\sqrt{2}} \leq hk \leq \sqrt{2}.$$

This is possible because consecutive dyadic numbers differ by a factor of 2. With $s = hk$, we have

$$\frac{M^{k-1}}{\varepsilon k} = e^{-h} \frac{h e^s}{\varepsilon s}.$$

As $\varepsilon \rightarrow 0$, $e^{-h} h/\varepsilon \rightarrow 1$. Since e^s/s decreases on $(0, 1]$ and increases on $[1, \infty)$,

$$\limsup_{\varepsilon \rightarrow 0} K_\varepsilon \leq \max_{1/\sqrt{2} \leq s \leq \sqrt{2}} \frac{e^s}{s} = \frac{e^{\sqrt{2}}}{\sqrt{2}}.$$

This proves the theorem.

Verification notes

The companion script `code/check_midpoint_bounds.py` checks the singular-value identity for the shear matrix and prints the numerical upper constant

$$\frac{e^{\sqrt{2}}}{\sqrt{2}} \approx 2.9085072356.$$

Novelty and literature check

The cheap run indexes were searched for arXiv:1105.0854, perturbations of isometries, Hyers-Ulam stability, midpoint constants, μ -isometries, Banach-Stone constants, and related phrases. No direct duplicate was found. Web searches for the exact title, K_ε , the midpoint expression, Gorak, Vestfrid, and the Banach-Stone $M_0 = \sqrt{2}$ problem found the source paper and adjacent perturbation literature, but no later answer to the midpoint constant in this bounded pass.

Limitations

This is not a full solution of the source problem. It improves the explicit interval for K from

$$0.5 \leq K \leq 3$$

to

$$1 \leq K \leq e^{\sqrt{2}}/\sqrt{2},$$

but it does not determine the exact value of K . It also does not address the separate final-remarks problem about the optimal Banach-Stone constant M_0 .

Human-review recommendation

This packet is a likely valid partial result. Review should focus on the global Lipschitz estimate for the nonlinear shear and on the dyadic choice in the upper-bound optimization.

References

- [1] R. Gorak, *Perturbations of isometries between Banach spaces*, arXiv:1105.0854.
- [2] I. Vestfrid, *Affine properties and injectivity of quasi isometries*, Israel Journal of Mathematics 141 (2004), 185–210.